MAHATWA KUMAR

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EDUCATION

University of California, Los Angeles (UCLA) — Los Angeles, CA

B.S. Mechanical Engineering, Samueli School of Engineering | ***GPA: 3.61/4.00*** — **Expected June 2028**

- **Coursework:** Fluid Mechanics, Intermediate Fluid Mechanics, Transport Phenomena, Thermodynamics, Dynamics, Materials Science, Dynamic Systems Lab, Circuit Measurements Lab, Computer-Aided Design

EXPERIENCE

Indian Institute of Technology Delhi — New Delhi, India

ASPIRE Summer Research Fellow, Mechanics of Soft Active Materials Lab (Prof. Sushma Santapuri) — **Summer 2026**

- Modeled an ultra-thin honeycomb underwater acoustic metamaterial in Fusion 360 (rubber–steel–rubber resonator cells), reproducing a published design for prototyping
- Adapted it for in-lab fabrication: 3D-printed resin in place of steel, and silicone elastomers selected via Shore-hardness conversion to hit a $\sim$0.5 MPa target modulus
- Modeled exosuit cable friction with the capstan equation to compare sheath routings and actuator placements

CO-OPTS — Remote

Engineering Design Intern — **Summer 2025**

- Took a modular product prototype in Fusion 360 from concept to final design, meeting all core functional requirements

University of California, Los Angeles — Los Angeles, CA

Course Grader, Thermodynamics (Prof. Wei) — **Spring 2026**

- Graded homework and exams for 102 students in upper-division Thermodynamics, with written feedback on methods

PROJECTS

F1 Following-Car Time-Loss Model | ***Python, OpenF1, lap simulation*** — **2026**

- Built a physics model of the lap time lost when following another car, splitting it into wake loss, close-following cost and traffic hold-up inside a quasi-steady lap simulation, with six global parameters and no per-circuit tuning
- Calibrated on $\sim$589,000 OpenF1 segment passes (2022–2025); leave-one-circuit-out, removed 65% of baseline error (15/16 circuits pass), vs 33% using FIA CFD wake data and 60% for pure ML
- On 7 sealed held-out circuits, 5 of 7 pass (57% of error removed); gain over the FIA-wake baseline holds under bootstrap (95% CI 0.29–0.32)
- Derived a "stall gap" where per-lap loss equals the follower's pace advantage; tested against 2,339 observed race plateaus, predicting gaps to 0.29–0.37 s error versus 0.44–0.47 s for a constant guess

F1 Car External Aerodynamics CFD | ***Ansys Fluent, Fluent Meshing, CadQuery*** — **2026**

- Modeled a 59-part ground-effect F1-style car in CadQuery; meshed it in Ansys Fluent Meshing (781k-cell poly-hexcore, prism layers, underfloor refinement) and solved k-ω SST RANS in Fluent with moving ground and rotating wheels
- Traced an 87%-front aero balance to a weak floor using a per-component force breakdown; a redesigned venturi floor raised downforce 1.9$\times$ ($C_L A$ 1.66$\rightarrow$3.21 m^2) at equal drag and moved balance to 62% front

EV Aerodynamics for Range | ***Ansys Fluent, CadQuery, Python*** — **2026**

- Ran 25 Ansys Fluent RANS simulations of an EV and 11 shape variants (0.32–0.86M cells, mesh independence within 0.6%), feeding a road-load range model over steady speeds and the EPA HWFET cycle
- Found that a smooth floor, 5° diffuser, wheel covers, lower ride height and camera mirrors cut drag area 37%, worth +137 km per charge at 120 km/h

EV Battery Cold Plate: CFD and Bench Rig | ***Ansys Fluent, SolidWorks, ESP32/C++*** — **2026**

- Compared serpentine, parallel and pin-fin cold plates by hand calculation and Ansys Fluent heat transfer CFD (36 operating points); predicted pin-fin matches serpentine cooling at 1/25 the pressure drop
- Designed the plates in SolidWorks (drawings, print files) and a $92 ESP32 FreeRTOS test rig (8 NTC channels at $\pm$0.1 K, latched safety trips) for ASME V&V 20 validation, build in progress

Gesture-Controlled Assistive Elbow Exosuit | ***SolidWorks, ESP32-S3, nRF52840*** — **2026**

- Designing a hands-free, cable-driven elbow orthosis armed by teeth taps and driven by head tilt (ear-worn IMU over BLE); sized the drive from a dynamics model (6.3 N·m peak) and modeled the 41-part assembly in SolidWorks

TECHNICAL SKILLS

- **CFD & Simulation:** Ansys Fluent & Fluent Meshing (k-ω SST, heat transfer), COMSOL, lap-time simulation
- **CAD & Prototyping:** SolidWorks, Fusion 360 (CAD/CAM), CadQuery (Python CAD); FDM, SLA and SLS 3D printing
- **Programming & Electronics:** Python, MATLAB, C++, embedded C (ESP32, nRF52), motor control, LaTeX

PUBLICATIONS & HONORS

- Kumar, M. & Bowal, K. (2021). Exoskeleton for Knee Arthritis Patients. *Journal of Student Research*, 10. DOI
- UCLA Samueli School of Engineering Scholarship Supplement • ASPIRE Summer Research Fellowship, IIT Delhi